



— 宏基舜达 —
H.J SHUNDA

STRUCTURAL STEEL vs REINFORCED CONCRETE

20-year parameter matrix - AISC 360-16 vs ACI 318

10,000 m² (107,640 sq ft) logistics hub



1 DEAD LOAD

RCC 180 kg/m² (36.9 lb/ft²)



Steel 32 kg/m² (6.6 lb/ft²)

Footing concrete 0.22 to 0.09 m³/m²

Excavation 2.5-3.5 m to 1.2-1.8 m
(8.2-11.5 ft to 3.9-5.9 ft)

Concrete 10-14 months



Steel 4-6 months



USD 4.50/m²/month: 5 months early = USD 225,000

3 SPAN

RCC: columns beyond 24 m (78.7 ft)

Steel clear span 30 / 36 / 48 m

(98.4 / 118.1 / 157.5 ft)

+8-12% usable storage volume, same footprint



4 MAINTENANCE

ISO 12944 C3, 160 um (6.3 mils): 15-20 years

ISO 1461 galvanising, 85 um: 25+ years

Concrete: floor joints re-caulked every 3-5 years

CONCRETE WINS THE SLAB. ABOVE THE SLAB, THE NUMBERS FLIP.

Is your tender scored on first cost -
or 20-year cost?



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